Course Title	OBJECT ORIENTED PROGRAMMING WITH JAVA		
Course Code	22CS307A	L-T-P-C	(2-0-1)3
Exam Hrs.	3	Hours / Week	3
SEE	50 Marks	Total Hours	40

Course Objective: Design and develop java application programs using object-oriented concepts.
Course Outcomes (COs): Upon completion of the course, students shall be able to:

#	Course Outcomes	Mapping to POs	Mapping to PSOs
1.	Apply Java constructs for the development of object oriented programs.	1	2
2.	Analyze the given java program to make suitable changes.	2	-
3.	Design a java program for the given problem.	3	2
4.	Conduct practical experiments for demonstrating object oriented concepts through java using IDE.	1, 2, 5	1

Course Contents:

MODULE – 1	13 Hrs

Java Programming Fundamentals – The Java Language, The key attributes of Object Oriented programming, Java Development Kit (JDK), JVM, A first Simple program, handling syntax errors, The Java Keywords, Identifiers in Java. **Introducing Data Types and Operators :** Java's Primitive Types, A Closer Look at Variables, The Scope and Lifetime of Variables, operators, Shorthand Assignments, Type conversion in Assignments, Using Cast, Operator Precedence, Expressions. Program Control Statements: Input characters from the Keyboard, if statement, Nested ifs, if-else-if Ladder, Switch Statement, Nested switch statements, for Loop, Enhanced for Loop, While Loop, do-while Loop, Use break, Use continue, Nested Loops. **Introducing Classes, Objects and Methods:** Class Fundamentals, How Objects are Created, Reference Variables and Assignment, Methods, Returning from a Method, Returning Value, Using Parameters, Constructors, Parameterized Constructors, The new operator Revisited, Garbage Collection. The this Keyword. More Data Types and Operators: Arrays, Multidimensional Arrays, Alternative Array Declaration Syntax, Assigning Array References, Using the Length Member, The For-each Style for Loop, Strings, The Bitwise operators.

MODULE – 2	12 Hrs

A Closer Look at Methods and Classes: Controlling Access to Class Members, Pass Objects to Methods, How Arguments are passed, Returning Objects, Method Overloading, Overloading Constructors, Recursion, Understanding Static, Introducing Nested and Inner Classes, Varargs: Variable-Length Arguments. **Inheritance:** Inheritance Basics, Member Access and Inheritance, Constructors and inheritance, Using super to C all Superclass constructors, Using super to Access Superclass Members, Creating a Multilevel Hierarchy, When are Constructors Executed, Superclass References and Subclass Objects, Method Overriding, Overridden Methods support polymorphism, Why overridden Methods, Using Abstract Classes, Using final, The object class.

MODULE – 3	13 Hrs

Interfaces: Interface Fundamentals, Creating an Interface, Implementing an Interface, Using Interface References, Implementing Multiple Interfaces, Constants in Interfaces, Interfaces can be extended, Nested Interfaces. Packages: Package Fundamentals, Packages and Member Access, Importing Packages, Static import. **Exception Handling:** The Exception Hierarchy, Exception Handling Fundamentals, The Consequences of an Uncaught Exception, using Multiple catch clauses, Catching subclass Exceptions, try blocks can be nested, Throwing an Exception, A Closer look at Throwable, using finally, using throws, Java's Built-in Exceptions.

MODULE – 4	12 Hrs

Multithreaded Programming: Multithreading fundamentals, The Thread Class and Runnable Interface,Creating Thread, Creating Multiple Threads, Determining When a Thread Ends, Thread Priorities, Synchronization, using Synchronization Methods, The Synchronized Statement, Thread Communication using notify(), wait() and notify All(), suspending, Resuming and stopping Threads. Exploring java.io – The I/O Classes and Interfaces, File, Auto closeable, Closeable and Flushable Interfaces, I/O Exceptions, The Stream Classes, The Byte Streams, The Character Streams, The Console Class.

Text Books :

1. Java Fundamentals, A comprehensive Introduction by Herbert Schildt, Dale Skrien.Tata McGrawHill Edition 2013 (Chapters 1,2,3,4,5,6,7,8,9,10,12)
2. Java –The complete Reference, by Herbert Schildt Eight Edition Tata Mcgraw Hill Education (Chapter 19).

Reference Books:

1. Programming in JAVA2 by Dr K Somasundaram ,Jaico publications
2. Java Programming by Hari Mohan Pandey, Pearson Education, 2012.
3. Deitel and Deitel – "Java How to Program" – 6th Ed. – Pearson.

MOOCs:

1. http://nptel.ac.in/courses/106106147/
2. http://www.nptelvideos.com/java/java_video_lectures_tutorials.php
3. https://www.youtube.com/watch?v=0KL_zftem4g
4. https://www.coursera.org/specializations/object-oriented-programming

Activity:

Write and execute the following programs in java

1. Design Java application to model Employee of an "ABC" organization. Consider types of employees as : a) Manager b) Sales Person Perform the following:
    - Implement simple inheritance where Employee (employee ID, First Name, Last Name, Current salary) is super class, consider Manager (number of stock options) and SalesPerson (number of sales, commission rate) as subclasses.
    - Define parameterized constructors in class hierarchy.
    - Define methods to display status of employee (override toString()) and increase current payment (salary) as giving bonus by a small amount.

2. Create class Employee with name, Id, cellNum, designation, department, salary permanent (boolean) as members. Also create separate class for member name as EmpName, department as Department and salary as Salary. Members of EmpName are firstName, midName, lastName . Members of Salary are BASIC, DA percent, HRA , PF, insurance. Members of Department are deptName, deptCode. Create an instance of Employee and initialize all its members from user input console. Access the same instance by more than one reference of Employee. Count the number of references and objects being created. [class, reference, objects].

3. Create a super class Bind {x,y} and subclass Bind3d {z} include
    - Parameterized constructor in class Bind to initialize object.
    - Include a method in class Bind to Print data members of object. Override the same method in sub¬class and reuse the inherited method in it.
    - Create final static member printArr (Bind[ ]) which print array of objects. In this method use above method to print member of individual objects of array.
    - Write test class to create to array of Bind3d type. Use the above method to print all theobjects of array.

4. A bank maintains two kinds of accounts – Savings account and Current account. The savings account provides simple interest, deposit and withdrawal facilities. The current account only provides deposit and withdrawal facilities. Current account holders should also maintain minimum balance. If balance falls below minimum level a service charge is imposed. Create an abstract class Account that stores customer name, account number type of account and abstract methods. From this derive the classes Curr_Account (double balance, double min_bal, double service Charge / penalty) and Sav_Account (double balance).

   Include the necessary methods in order to achieve the following:

   Define parameterized constructor in a class hierarchy.

   - Allow deposit and update the balance.
   - Display the balance.
   - Compute interest and add to balance.
   - Permit withdrawal and update the balance (check for minimum balance).
   - Apply polymorphism if required for methods in class hierarchy.
   - Create an array of super class / object and populate with subclass objects and call the overridden /object methods.
   - Write a test program to demonstrate the above said implementations.

5. Define an interface EMP Interface (void displayEMP(), void giveBonus (double amount)). Define an abstract class Employee(empID, fName, lName, salary). Define a concrete class Manager (noOfStockOptions), subclass of Employee and define interface methods.

   Perform the following:

   - Define appropriate constructors in a class hierarchy.
   - Ensure the bonus amount should not be negative and zero using exception handling mechanism (use throws and throw clauses of exception handling)
   - Create array of interface reference variables and populate with manager objects.

   Write a test program to implement the above said requirements of interface implementation and exception handling.